



Fashion Retail Customer Behavior Analysis: Unlocking Sales & Loyalty Drivers

Role: Data Analyst | Tools: Python (Pandas), SQL (PostgreSQL), Power BI

Dataset: 3,900 Transaction Records | Domain: Retail & E-commerce

by Humaira Zeanova

Executive Summary

This project optimizes sales and customer retention for a retail company by analyzing 3,900 transaction records. Key findings include a significant "gender gap" in revenue and inefficiencies in the subscription program.



Male Revenue Dominance

Men contribute 68% of total revenue (\$157,890), significantly outperforming women.



Subscription Paradox

Subscribers (\$59.49 avg spend) spend slightly less than non-subscribers (\$59.87 avg spend).

Strategic recommendations focus on restructuring subscription benefits and gender-targeted marketing to boost Average Order Value (AOV) by 15-20%.



Business Problem Statement

Retail management seeks to understand consumer decision drivers (discounts, ratings, seasons) to enhance long-term loyalty. The core question is: "How can the company leverage consumer shopping data to identify trends, improve engagement, and optimize marketing strategies?"

Revenue Drivers

Identify the most profitable demographic segments and products.

Loyalty Evaluation

Measure the effectiveness of the subscription program on total spending.

Customer Segmentation

Group customers for more personalized marketing strategies.

Data Preparation & Cleaning (Python)

Raw data was processed using Python (Pandas) to ensure quality before analysis.

01

Handling Missing Values

37 missing values in 'Review Rating' were imputed using the median rating per product category.

02

Redundancy Check

'Promo Code Used' was removed due to 100% correlation with 'Discount Applied'.

03

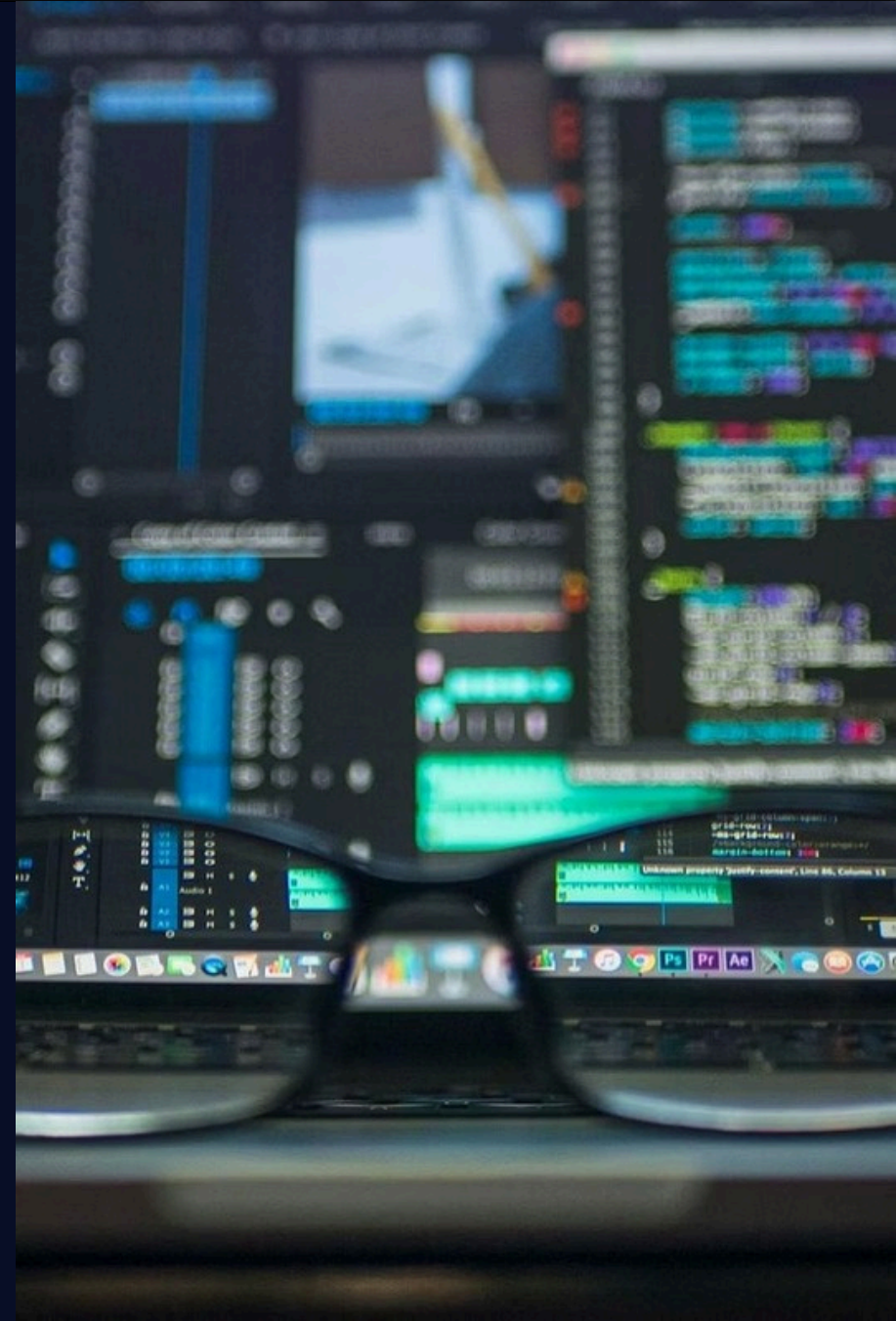
Feature Engineering

Created 'age_group' using quartile binning and converted 'Frequency of Purchases' to numerical 'purchase_frequency_days'.

04

Standardization

Column names were converted to 'snake_case' for SQL compatibility.



Key Insights & Analysis (SQL)

SQL queries revealed critical findings from the cleaned data.

A. The "Gender Gap" Revenue

Contrary to common retail assumptions, male customers are the largest revenue contributors.

Male	\$157,890	67.7%
Female	\$75,191	32.3%

Insight: Men are high-value customers, suggesting potential bias in current marketing or higher average product prices for men.



The Subscription Paradox & Customer Loyalty

B. The Subscription Paradox

Loyalty programs typically boost spending, but data shows an anomaly:

- **Subscribers:** 1,053 customers (27%) → Avg Spend: \$59.49
- **Non-Subscribers:** 2,847 customers (73%) → Avg Spend: \$59.87

Insight: Current subscription incentives do not encourage higher spending per transaction, possibly attracting only discount seekers.

C. Customer Loyalty Segmentation

Segmentation based on previous purchases reveals a mature customer base:

- **Loyal (>10 purchases):** 3,116 customers (~80%)
- **Returning (2-10 purchases):** 701 customers
- **New (1 purchase):** 83 customers

Insight: The primary focus should be on retention and upselling to existing loyal customers, rather than new acquisition.



Product & Shipping Performance



Top Category

Clothing dominates with items like Blouse and Pants (171 sold each).



Shipping Impact

Express Shipping users spend more (\$60.48 avg) than Standard (\$58.46 avg), indicating less price sensitivity.



Top-Rated Product

Gloves (3.86), Sandals (3.84), and Boots (3.82) have the highest average review ratings.



Discount-Dependent Products

Hat (50%), Sneakers (49.66%), and Coat (49.07%) have the highest percentage of discounted purchases.

Strategic Recommendations

Based on the data, here are actionable business recommendations:



Revamp Subscription Program

Introduce tiered benefits, e.g., "Free Express Shipping" for purchases over \$75 to increase AOV from \$59 to \$75.



Male-Centric Marketing

Allocate 60% of the advertising budget to male audiences. Create product bundles like "Essential Men's Look" (Jeans + Sweater) to boost transaction value.



Loyalty Program for "Super Loyal"

Shift from general discounts to exclusive access (e.g., early access to new seasonal products) for the 80% loyal customer base to maintain exclusivity and prevent churn.

Technical Appendix: Python & SQL

A. Python: Data Cleaning & Feature Engineering

```
# Impute Missing Values
df['Review Rating'] = df.groupby('Category')['Review
Rating'].transform(lambda x: x.fillna(x.median()))

# Drop Redundant Column
df = df.drop('Promo Code Used', axis=1)

# Binning Customer Age
labels = ['Young Adult', 'Adult', 'Middle-Aged', 'Senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels=labels)
```

B. SQL: Analysis Queries

Query 1: Revenue by Gender

```
SELECT gender, SUM(purchase_amount) as revenue
FROM customer
GROUP BY gender;
```

Query 2: Subscriber Spending Analysis

```
SELECT subscription_status, COUNT(DISTINCT customer_id) as
total_customers, ROUND(AVG(purchase_amount), 2) as avg_spend
FROM customer
GROUP BY subscription_status;
```

Query 3: Customer Segmentation

```
WITH customer_type AS (
  SELECT customer_id, previous_purchases,
  CASE
    WHEN previous_purchases = 1 THEN 'New'
    WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
    ELSE 'Loyal'
  END AS customer_segment
  FROM customer
)
SELECT customer_segment, COUNT(DISTINCT customer_id) as
total_customer
FROM customer_type
GROUP BY customer_segment;
```

Power BI Dashboard

An interactive Power BI dashboard, "Fashion Retail Customer Analytics Dashboard", visually presents key insights for data-driven decisions.

3.9K
Customers

\$59.76
Avg Purchase

3.75
Avg Review

The dashboard includes filters for Subscription Status, Gender, Category, and Shipping Type, along with visualizations for revenue, sales, and customer demographics. Take a look at the dashboard preview >>

View Full Project Repository

zeanovaa/
customer_behavior_proje...



End-to-end analysis of customer shopping behavior (3,900+ transactions) using Python, SQL, and Power BI. Uncovered key insights on gender revenue...

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Contributor

0

Issues

0

Stars

0

Forks



 GitHub



GitHub-zeanovaa/customer_behavior_project: End-to-end analysis of custome...

End-to-end analysis of customer shopping behavior (3,900+ transactions) using Python, SQL, and Power BI. Uncovered key insights on gender revenue drivers and subscription model...

Fashion Retail Customer Analytics Dashboard

Subscription Status

No

Yes

Gender

Female

Male

Category

Accessories

Clothing

Footwear

Outerwear

Shipping Type

☐ 2-Day Shipping

☐ Express

☐ Free Shipping

☐ Next Day Air

☐ Standard

☐ Store Pickup

3.90K

Number of Customers

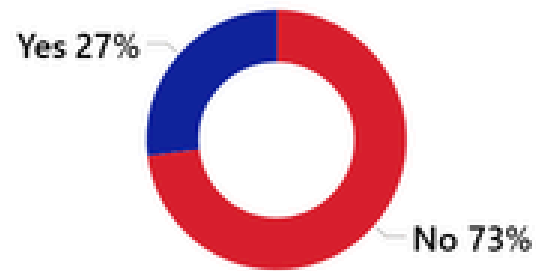
3.75

Average Review Rating

\$59.76

Average Purchase Amount

% of Customers by Subscription



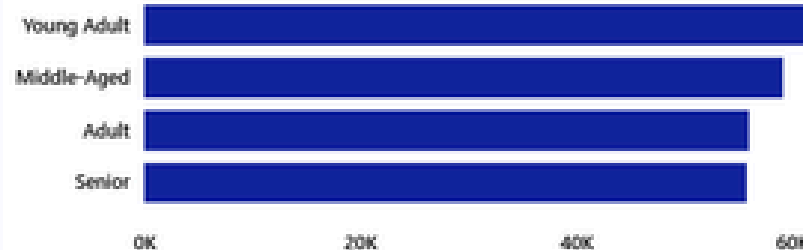
Revenue by Category



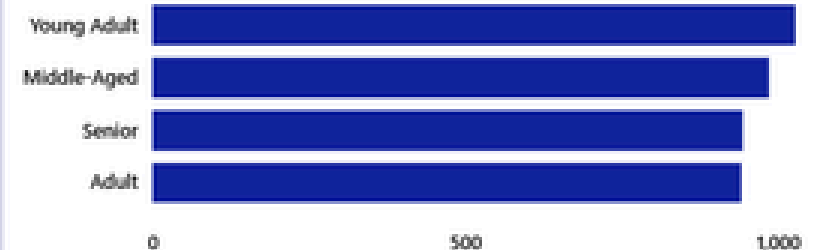
Sales by Category



Revenue by Age Group



Sales by Age Group



THANK YOU!

Let's Connect

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Credits & Acknowledgements

This project was developed as part of a comprehensive data analysis study.

Resource: Amlan Mohanty

Provided the dataset and end-to-end project road map.

Executed independently using Python (Jupyter Notebook),

SQL (PostgreSQL), and Power BI by Humaira Zeanova.